PS 4 (2)
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Productivity Model Specification

$$\log y_{it} = \alpha_0 + \alpha_1 \log y_{it-1} + \alpha_2 \log y_{it-2} + \alpha_3 \exp_{it} + \alpha_4 \exp_{it}^2 + \alpha_5 rank_i + \alpha_6 sex_t + \alpha_7 d80_i + \eta_i + \varepsilon_{it}$$

y_{t-1}, y_{t-2}: productivity in the previous 2 periods
exp_{it}: experience in academia measured current year-phd year
exp_{it}^2: experience^2
$rank_i$: $1/ ((t-t_{0i})r_i)$
sex_i: male=0
$d80_i$: if one gets a Phd degree after 1980 then d80 =1
η_i: individual effect

Productivity Model Estimation

$$\log y_{it} = \alpha_0 + \alpha_1 \log y_{it-1} + \alpha_2 \log y_{it-2} + \alpha_3 \exp_{it} + \alpha_4 \exp_{it}^2 + \alpha_5 rank_i + \alpha_6 sex_t + \alpha_7 d80_i + \eta_i + \varepsilon_{it}$$

$logy_{it-1}$, $logy_{it-2}$: endogenous time variant variables X_2
$exp_{it,}$ exp_{it}^2: exogenous time variant variables X_1
$rank_i$: endogenous time invariant variable Z_2
sex_i, $d80_i$: exogenous time invariant variable Z_1
η_i: individual effect

HTIV instruments: $Pexp_{it}$, $Pexp_{it}^2$, $Qexp_{it}$, $Qexp_{it}^2$, $Qlogy_{it-1}$, $Qlogy_{it-2}$, sex_i, $d80_i$

Productivity Model Estimation

$$\log y_{it} = \alpha_0 + \alpha_1 \log y_{it-1} + \alpha_2 \log y_{it-2} + \alpha_3 \exp_{it} + \alpha_4 \exp_{it}^2 + \alpha_5 rank_i + \alpha_6 sex_i + \alpha_7 d80_i + \eta_i + \varepsilon_{it}$$

1st step: 2SLS use $Pexp_{it}$, $Pexp_{it}^2$, $Qexp_{it}$, $Qexp_{it}^2$, $Qlogy_{it-1}$, $Qlogy_{it-2}$, sex_i, $d80_i$ as instruments , get Prsq

2nd step: Regress Prsq against 1/T and constant, get estimated σ_u^2 and σ_a^2, construct $\theta = \sigma_u/(\sigma_u^2 + T\sigma_\alpha^2)^{1/2}$

3rd step: replace every variable with $y - (1-\theta)\bar{y}$, run 2SLS again

Productivity Model Results

$$\log y_{it} = \alpha_0 + \alpha_1 \log y_{it-1} + \alpha_2 \log y_{it-2} + \alpha_3 \exp_{it} + \alpha_4 \exp_{it}^2 + \alpha_5 rank_i + \alpha_6 sex_t + \alpha_7 d80_i + \eta_i + \varepsilon_{it}$$

Source	SS	df	MS
Model	40192.0489	8	5024.00612
Residual	1937.19777	5440	.356102531
Total	42129.2467	5448	7.7329748

Number of obs	=	5448
F(8, 5440)	=	.
Prob > F	=	.
R-squared	=	.
Adj R-squared	=	.
Root MSE	=	.59674

y	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
y1	-.0664904	.0123242	-5.40	0.000	-.0906508	-.0423301
y2	-.0460426	.012356	-3.73	0.000	-.0702654	-.0218198
exp	.1620468	.0054651	29.65	0.000	.151333	.1727607
expsq	-.0033489	.0001342	-24.96	0.000	-.0036119	-.0030859
ier	1.425002	.292866	4.87	0.000	.8508669	1.999136
d80	-.0251034	.0213236	-1.18	0.239	-.0669062	.0166994
sex	-.0047895	.0282889	-0.17	0.866	-.060247	.0506681
theta	1.988024	.0830962	23.92	0.000	1.825122	2.150926